

VIRTUAL COMMUNITY: AN OPEN WORLD FOR HUMANS, ROBOTS, AND SOCIETY

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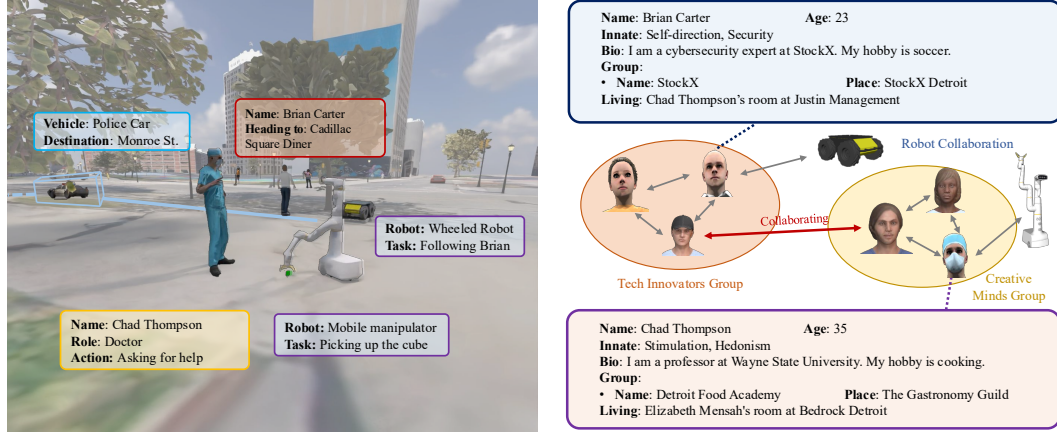


Figure 1: **Virtual Community supports embodied multi-agent simulation in open-world environments.** We introduce an automated pipeline that generates open-world scenes and agent communities, with agents instantiated as humanoid avatars or robots to enable diverse social interactions.

ABSTRACT

The rapid progress of AI and robotics may profoundly transform society, as humans and robots begin to coexist in shared communities, bringing both opportunities and challenges. To explore this future, we present Virtual Community—an open-world platform for humans, robots, and society—built on a universal physics engine and grounded in real-world 3D scenes. To support these, Virtual Community features: 1) An open-source multi-agent physics simulator that supports robot, human, and their interactions within a society; 2) A large-scale, real-world aligned pipeline, including vast outdoor space, diverse indoor scenes, and a community of grounded agents with rich characters and appearances. Leveraging Virtual Community, we propose two novel challenges. The *Community Planning Challenge* evaluates multi-agent reasoning and planning in open-world settings, such as cooperating to help agents with daily activities and efficiently connecting other agents. The *Community Robot Challenge* requires multiple heterogeneous robots to collaborate in solving complex open-world tasks. We evaluate various baselines and demonstrate the challenges in both high-level open-world task planning and low-level cooperation controls. We have open-sourced our project* and hope that Virtual Community will unlock further study of human-robot coexistence in open worlds.

1 INTRODUCTION

In recent years, the development of intelligent embodied agents has been propelled by advances in virtual simulators (Savva et al., 2019; Kolve et al., 2017; Anderson et al., 2018; Todorov et al., 2012;

* Anonymous codebase: <https://virtual-community-ai-25.github.io/>

Savva et al., 2017; Li et al., 2021; Xiang et al., 2020b; Makoviychuk et al., 2021; Gan et al., 2021; Cheng et al., 2024; Wu et al., 2024; Wang et al., 2024a; Li et al., 2024; Zhong et al., 2024; Zhuang* et al., 2025; Du et al., 2019). However, most of these platforms focus on robots (Tao et al., 2024; Xiang et al., 2020b; Li et al., 2024), human-like agents (Puig et al., 2018; 2020), or only a limited number of agents with simple interactions (Puig et al., 2023; Gan et al., 2021). In contrast, support for large, heterogeneous communities of human and robot agents in scalable open worlds remains limited, constraining the study of complex multi-agent behaviors between humans and robots.

To address this challenge, simulators must support the following key features. First, they should offer physically realistic simulations that accommodate large communities of human-like avatars and robots. Existing multi-agent embodied AI platforms (Puig et al., 2018; 2023; Gan et al., 2021; Li et al., 2021; Wu et al., 2024) typically handle only small groups of avatars or robots, or provide limited physics-based interactions, thereby constraining the realism of community-level behaviors. Second, the simulator must support the creation of diverse, populated worlds, including large-scale 3D environments and scene-grounded agent communities. Current approaches fall into two categories: manual design or procedural generation (Wang et al., 2024a; Gan et al., 2021; Tsoi et al., 2022; Wu et al., 2024; Gao et al., 2024), which enable rich agent–environment interactions but suffer from limited diversity and realism; and 3D reconstruction methods (Savva et al., 2019; Shen et al., 2021), which produce visually realistic and varied scenes but require extensive visual input and often yield low-interactivity environments in open-world settings.

In this paper, we present Virtual Community, an open world for humans, robots, and society. Virtual Community addresses these challenges by building a unified simulation framework for human-like agents and based on the Genesis (Authors, 2024) physics engine and integrating large-scale, real-world geospatial data with generative models to produce interactive, scalable open worlds (Figure 1). The platform offers two key advancements:

Unified Simulation for Avatars and Robots Virtual Community simulates human-like avatars and diverse robots within the generated open worlds using a unified framework based on the Genesis (Authors, 2024) physics engine, supporting diverse physical and social interactions among different types of agents. Virtual Community also provides agents with a unified interface with distinct observation and action spaces.

Virtual Community fully automates the generation of open worlds with several key features: (1) scalable, real-world-aligned outdoor scenes of customizable size and quantity, along with corresponding indoor scenes and annotations; and (2) generation of agent communities endowed with scene-grounded profiles and social relationship networks. Virtual Community combines generative models with real-world geospatial data, ensuring scalability in data volume, realism, and extent.

Virtual Community enables a variety of new possibilities in embodied AI research. The expansive open-world scenes and their agent communities introduce a new challenge of multi-agent task planning in open worlds. We introduce the *Community Planning* challenge as a first step in this direction. This challenge includes assistant tasks, in which to provide assistance in daily open world activities, and social influence tasks, in which agents must efficiently explore the community and connect with one another. Virtual Community also supports physically realistic simulations of interactions, for which we propose the *Community Robot* challenge. This challenge tasks with cooperating to complete tasks that involve both indoor and dynamic open-world environments.

Our simulator advances the field by enabling unified simulations of human and robot communities in generated open worlds, surpassing existing solutions in both scope and capability. By overcoming limitations in the scalable simulation of humans, robots, and societies, we pave the way for studying embodied general intelligence that can coexist with complex, interconnected human communities.

2 RELATED WORKS

Embodied AI Simulation Recently, embodied AI has seen significant advancements through the development of simulation platforms. Most existing simulators primarily focus on household tasks within indoor scenes (Beattie et al., 2016; Savva et al., 2019; Yi et al., 2018; Das et al., 2018; Xiang et al., 2020a; Shen et al., 2021; Li et al., 2021; Puig et al., 2018; Kolve et al., 2017; Yan et al., 2018; Li et al., 2024; Tao et al., 2024; Deitke et al., 2020; 2022b), while some have extended support to outdoor environments (Gan et al., 2021; Tsoi et al., 2022; Wang et al., 2024a; Dosovitskiy

et al., 2017; Kendall et al., 2018; Gulino et al., 2023; Wu et al., 2024). However, existing platforms lack the diverse and scalable outdoor environments needed to support larger agent populations and more complex multi-agent interactions. In contrast, this paper introduces a simulation platform with expansive open-world environments, integrating both indoor and scalable outdoor scenes to facilitate broader agent interactions and enable more intricate task scenarios.

Embodied Social Intelligence Current research on *Embodied Social Intelligence* is often limited to small agent populations in constrained household scenarios (Puig et al., 2020; Zhang et al., 2023; Stone et al., 2022; Savva et al., 2019; Jain et al., 2020; Szot et al., 2023; Zhang et al., 2024) or simplified to 2D or grid worlds (Carroll et al., 2019; Suarez et al., 2019; Tsoi et al., 2020; Samvelyan et al., 2019; Yu et al., 2024; Yang et al., 2024a), hindering model development in the open world. Specifically, (Park et al., 2023) demonstrates the robust simulation of human-like agents within a symbolic community, ignoring the 3D perception and realistic physics in the open world. (Wang et al., 2023c) studies human-like simulation guided by system 1 processing with basic needs. Predominant approaches, such as multi-agent reinforcement learning (MARL) and other planning models, face several limitations when applied to open-world settings. MARL, for instance, often struggles with scalability due to the exponential growth of state and action spaces as the number of agents increases (Wen et al., 2022). This makes it difficult to learn effective policies in complex, dynamic environments. Additionally, MARL approaches typically require extensive training data and computational resources, which may not be feasible in real-world applications. Other planning models, while potentially more efficient, often lack the adaptability required to handle the unpredictable nature of open-world interactions. They may rely on predefined rules or assumptions that do not hold in all scenarios, leading to suboptimal performance and limited generalization to new contexts (Puig et al., 2020).

Foundation and Generative models for Embodied AI With the recent advance of foundation models (Bubeck et al., 2023; Liu et al., 2023; Driess et al., 2023; Blattmann et al., 2023), numerous works have explored how they can help build powerful embodied agents (Wang et al., 2023b; Xi et al., 2023; Sumers et al., 2023; Wang et al., 2023d; Ahn et al., 2022; Sharma et al., 2021; Wang et al., 2023a; Park et al., 2023; Hong et al., 2024; Black et al., 2024b), and scenes for simulation (Yang et al., 2024c; Hu et al., 2024; Höllein et al., 2023; Schult et al., 2023; Deitke et al., 2022a; Fu et al., 2021; Yang et al., 2024b; Feng et al., 2024; Tang et al., 2023; Paschalidou et al., 2021; Shcherbyna et al., 2024; Deitke et al., 2023). RoboGen (Wang et al., 2024c) utilizes foundation models to automatically generate diverse tasks, scenes, and training supervision, scaling up robotic skill learning with minimal human input. In contrast, our work fully integrates a generative pipeline into the simulation platform to create expansive open-world scenes and agent communities.

3 GENERATING OPEN WORLDS FOR SIMULATION

3.1 SCALABLE 3D SCENE CREATION

The existing 3D geospatial data API^{*} provides extensive data in terms of quantity and diversity. However, they are not directly suitable for embodied AI research. First, these geospatial data often contain noise, including transient objects and unrealistically rugged terrain that can disrupt simulations. Second, visual quality is inadequate for ground-level agent perspectives because these environments are typically reconstructed from aerial imagery and lack visual detail at ground level.

To bridge this gap, we propose an online pipeline that performs comprehensive cleaning and enhancement in both geometry and texture to make the scenes suitable for embodied AI simulations. This pipeline consists of four steps: mesh simplification, texture refinement, object placement, and automatic annotation. The pipeline supports automatic creation of 3D urban scenes at arbitrary locations. Using this pipeline, we generated 35 annotated scenes of various cities worldwide and present some qualitative examples of these scenes in Figure 3.

Geometry Reconstruction and Simplification Since the mesh topologies in 3D geospatial data sources are unreliable for embodied AI simulations, we decompose scenes into terrain, building, and decorative-roof elements, then apply specialized reconstruction operations to each component to make the entire scene simulation-ready. The terrain is generated procedurally from sparse refer-

^{*}<https://www.google.com/maps/>

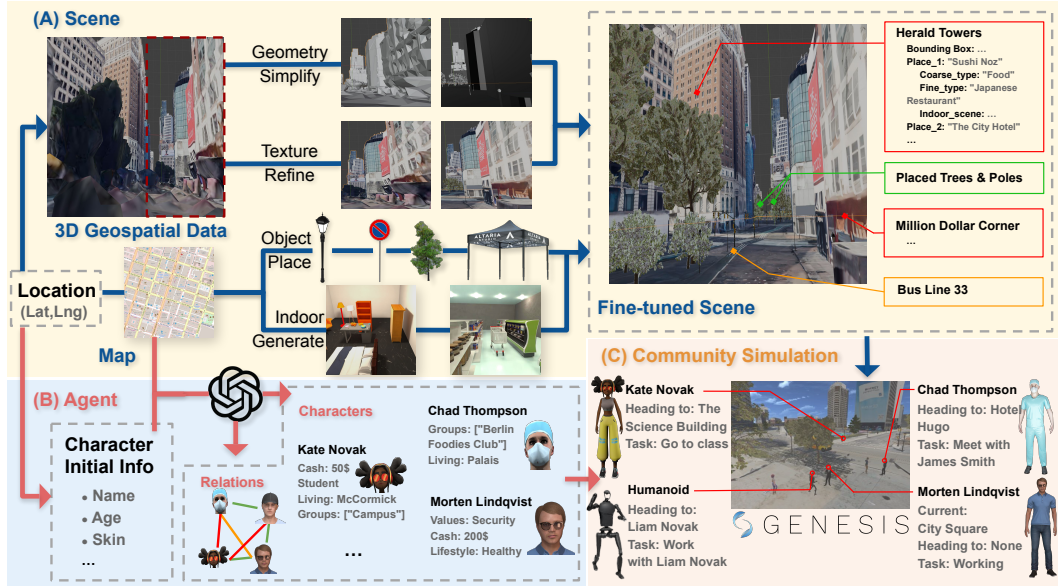


Figure 2: **Framework of the Virtual Community Generation Pipeline.** This pipeline generates scenes and corresponding agents from real-world geospatial data. The **scene generation** component (A) refines rough 3D data by using generative models to enhance textures and geospatial data to simplify geometry. It also utilizes generative methods to create interactive objects and detailed indoor scenes. The **agent generation** component (B) leverages LLMs to generate agent characters and social relationship networks based on scene descriptions. (C) We simulate the human communities and robots in the open world scenes based on Genesis engine.

ence elevation points via bilinear interpolation. We then derive simple, topologically sound building meshes using OpenStreetMap (OSM) data. Each building mesh is automatically adjusted to better match the Google 3D Tiles geometry and to align with the terrain elevation. By aligning mesh geometries to OSM primitives, we remove unnecessary details and artifacts—such as distorted surfaces and irregular shapes resulting from aerial reconstruction errors—thereby denoising the meshes for more efficient physics simulations and improved rendering performance.

Texture Enhancement for Realistic Simulation We further apply advanced image-processing techniques to enhance mesh textures. During mesh construction and simplification, textures from the original 3D Tiles are baked onto new geometries, which can result in missing or distorted regions. To address these issues, we first employ a Stable Diffusion 3 (Stability AI, 2024) based inpainting method to remove noise and repair damaged or incomplete textures. We then refine texture details using street-view imagery. This two-step process significantly improves visual fidelity, making textures more suitable for ground-level rendering.

Object Replacement for Interactive Scene To enhance scene interactivity, we combine generative and retrieval methods to populate the environment with interactive objects (e.g., bikes and tents). Using OSM annotations, we identify object types and locations to reflect real-world contexts. For relatively simple objects, such as tents, we adopt a generative pipeline that uses OSM text annotations on amenities as input: a Stable Diffusion model (Rombach et al., 2021) first generates images of the relevant objects, which are then processed by the One-2-3-45 framework (Liu et al., 2024) to produce corresponding 3D meshes. For more complex objects, such as trees, we use the retrieval pipeline, which randomly samples assets whose categories match the OSM annotations from a pre-collected dataset.

Place and Transit Annotations with Geospatial Data To facilitate alignment with real-world locations and provide semantic context for community activities, we developed a pipeline to automatically annotate places, buildings, and public transit within scenes using geospatial data. First, we query Google Maps Places for location information in the target area and organize results into different categories. Next, we use OSM to retrieve building names and bounding boxes, matching them



Figure 3: **Egocentric view of the generated scenes.** The resulting scene features clean geometry and realistic textures, which support physical simulation and enhance real-world style fidelity.

with the place entries. We then filter out unmatched or inaccessible locations to generate accurate place annotations. Finally, we annotate bus transit routes based on these place annotations. These metadata enable agents to access location-specific information and support tasks that require spatial context, such as navigation and location-based decision-making, and also power traffic simulation, including buses, pedestrians, and other vehicles.

Indoor Scenes Creation To create indoor scenes in the communities, we employ a pipeline combining generation and retrieval to produce detailed, realistic multi-room environments. The pipeline’s input is the building names in the target area, obtained from Google Maps and OSM. We first retrieve indoor layouts from GRUTopia (Wang et al., 2024a) for categories such as offices, restaurants, and stores. For building types not covered by GRUTopia, we use Architect (Wang et al., 2024b) to generate the corresponding indoor rooms for simulation.

3.2 AGENT COMMUNITY GENERATION

Given diverse generated scenes from real-world geospatial data, we introduce a generative pipeline to populate these environments with communities of agents endowed with grounded character profiles and social relationship networks, .

Characters and Social Network Generation We utilize the open-world knowledge of the Large Language Model (LLM) to generate agent character profiles and personalities grounded in the scene. Specifically, we use GPT-4o to perform this generation. The input to the LLM is structured into two parts to create characters grounded in a specific scene. The first part contains scene-related information, such as the scene name and details about various places, including their names, types, and functionalities. The second part includes details on the agents’ appearances to ensure consistency between their visual attributes and generated profiles, which are annotated with the name and age. With both parts provided, the LLM generates agent profiles along with their social relationships. The profiles consist of basic attributes such as names, ages, occupations, personalities, and hobbies, which influence each agent’s daily activities. Social relationships are structured as groups, each containing a subset of agents along with a text description and a designated place for group activities, connecting these agents into a cohesive community.

Grounding Validator We implemented a grounding validator that verifies whether generated character profiles are accurately grounded in the scene by ensuring all referenced places exist. If validation fails, the LLM receives feedback from the validator and attempts to correct the mismatch. Detailed examples of prompts used in the pipeline, generated characters, and social relationship networks are provided in the Appendix. K.

Human-Like Avatar Creation We first obtained 20 avatar skins representing diverse genders, professions, and appearances from Mixamo* for integration into the Virtual Community. We also used the Avatar SDK* to generate high-fidelity human meshes from synthetic human face images from FaceSynthetics (Wood et al., 2021), enabling representation of diverse individuals.

*<https://www.mixamo.com/>

*<https://avatarsdk.com>

3.3 UNIFIED SIMULATION FOR HUMAN AND ROBOT COMMUNITY

Virtual Community provides a unified framework for simulating both robots and human agents in the community. It implements an avatar simulation framework for human agents, while robot simulations are largely inherited from Genesis.

Avatar Simulation and Control To simulate avatars in Genesis physics engine, we combine SMPL-X human skeletons with these avatar skins to model human avatars. The motions of these avatars are parameterized by SMPL-X pose vectors $J \in \mathbb{R}^{162}$ and global translation and rotation vectors $T, R \in \mathbb{R}^3$. We use over 2,000 motion clips from Mixamo and adjust their playback speeds to match our avatars, including walking, object manipulation, and vehicle entry. For walking, we loop the clip until the avatar covers the required distance. For object-related actions, objects are kinematically attached to or detached from the avatars’ hands based on the action. Similarly, during vehicle-related motions, avatars are kinematically attached to or detached from vehicles. We also incorporate physics constraints: collision detection is performed between avatars and scene entities, and motion terminates upon detection of a potential collision.

Daily Schedule Generation and Simulation Given the scene-grounded character profiles and social relationship networks, we prompt foundation models to generate each agent’s daily schedule (Park et al., 2023). However, we structure each schedule so that every activity includes a start time, an end time, an activity description, and a corresponding location. We also explicitly account for the commute time between activities at different locations to reflect the actual cost of navigating an expansive 3D environment. This approach allows agents to follow the organized high-level plan effectively and maintain consistency over time. During simulation, agents follow the generated schedules to carry out daily activities. Examples of detailed daily plans are provided in the Appendix. [B.2](#).

Robot Agent Simulation We simulate robots alongside avatars in the Genesis simulator. Virtual Community supports five types of robots: drones, quadruped robots, humanoid robots, wheeled robots, and mobile manipulators, each with a distinct robot controller. The robot controller bridges the interface between Virtual Community and Genesis, exposing only selected action spaces. Virtual Community shares the same simulation loop between avatars and robots with different control frequencies. To support faster collision detection during robot physics simulation, we use an invisible terrain mesh and decomposed building meshes as collision geometry for the background scene, enabling more efficient physics simulation.

4 OPEN WORLD MULTI-AGENT PLANNING

Based on Virtual Community, we propose the *Community Planning Challenge* to evaluate multi-agent planning capabilities in outdoor and indoor environments. The challenge comprises three *community assistant* tasks, in which agents cooperatively plan to assist multiple humans with daily open-world activities, and a *community influence task*, in which agents competitively plan to efficiently connect and interact with other agents in the community to increase their social influence.

4.1 COMMUNITY ASSISTANT TASKS

The community assistant tasks include the following three categories that require agents to plan cooperatively to provide humans with assistance on daily activities:

- **Carry:** Locate people and follow them to help carry objects to their home.
- **Delivery:** Move objects from source locations (indoor or outdoor) to a destination.
- **Search:** Locate target objects within an outdoor region or an indoor room.

Task Settings Each task includes multiple subtasks. For example, the Carry task requires agents to help humans carry several objects while following them. Therefore, adaptive task planning is essential for scheduling these runs, routing in the dynamic open world between waypoints, and managing task-level dependencies. We study two settings with different numbers of assistants: the *1-assistant* setting, in which a single assistant needs to provide assistance to human agents in the community, and the *2-assistants* setting, in which two assistants plan cooperatively to provide assistance.

Table 1: **Main results of the Multi-Agent Community Planning Challenge.** We report Success Rate (SR), Time Consumed (Ts), and Human Following Rate (HR) for three community assistance tasks averaged over 24 scenes.

Method	Carry		Delivery		Search		Avg SR↑
	SR↑	HR↑	SR↑	Ts↓	SR↑	Ts↓	
1-assistant							
Random	0.0	0.0	0.0	1500.0	0.0	1500.0	0.0
Heuristic	34.7	16.5	46.5	1462.9	45.1	1440.3	42.1
MCTS Planner	42.3	7.7	39.6	1500.0	45.1	1500.0	42.4
LLM Planner	29.9	15.3	41.7	1500.0	70.1	1339.0	47.2
2-assistant							
Heuristic	52.8	25.6	59.7	1415.8	51.4	1364.3	54.6
MCTS Planner	42.4	8.0	43.8	1500.0	48.6	1500.0	44.9
LLM Planner	30.2	10.7	43.8	1500.0	77.8	1141.3	50.6

Observation and Action Spaces At each simulation step, agents are provided with an observation consisting of RGB-D images with the corresponding camera matrix, segmentations, current poses, and task information. The action space includes *move forward*, *turn left*, *turn right*, *enter/exit bus/bike*, and *communicate*. The movement and turning actions can be set with a continuous amount.

Evaluation Metrics We provide three evaluation metrics for all assistant tasks: success rate (SR), defined as the number of successful subtasks divided by the total number of subtasks; average time consumed (T) per task; and human following rate (HR) for the carry task, defined as the number of frames in which the agent follows a human within a specified distance range divided by the total number of frames during the task. When the simulation reaches a total step limit of 1500, it stops and the results are evaluated automatically.

Baselines

- **Perception and Navigation Module:** We implement all baseline agents within a hierarchical planner framework. All agents employ the same low-level navigation algorithm, which reconstructs a point cloud from egocentric RGB-D observations at each step and converts it into a volumetric grid representation at a resolution of 0.1 m. Based on this grid, a 2D occupancy map with a resolution of 0.5 m is generated, and an A* algorithm is used to efficiently compute the shortest path to a bounding box. To accommodate dynamic environments, the navigation module recalculates the optimal path at every step.

- **Random Planner:** The Random planner is a trivial planner that randomly selects from the space of high-level actions without any planning.

- **Heuristic Planner:** The heuristic planner is based on a finite-state automaton defined by domain experts. At each state, the agent takes an action such as navigating, picking objects, entering rooms (see details in Appendix F.4).

- **MCTS Planner:** We also introduce a new Monte-Carlo Tree Search (MCTS) based baseline planner, which employs Monte-Carlo Tree Search to optimize task plans.

- **LLM Planner:** We follow CoELA (Zhang et al., 2023) to design an LLM Planner with a modular framework driven by gpt-4o to generate and select subplans, including navigation to specified open-space locations, searching for objects, entering indoor areas, and performing object manipulation actions (pick and put). At each decision point, the LLM is prompted with the current state and task objectives and produces a subplan—a sequence of high-level actions (Song et al., 2022), which is then executed step by step by the agent. A communication module is also adopted to facilitate the cooperation among agents through natural language communication.

Results We evaluate the above baselines on all three tasks. The results in Table 1 demonstrate the challenges of planning in open-world environments. From our experiment, the *Random* baseline fails to understand spatial relationships among tasks and no single method prevails for all the tasks. One common failure mode of baseline agents is underestimating the cost of open-world navigation and search, which leads to suboptimal task arrangement. Heuristic Planner performs strongly in the delivery task. While the LLM Agent prevails by a large margin in the search task, which involves

Table 2:

Method	Carry		Delivery		Search		Avg SR↑
	SR↑	HR↑	SR↑	Ts↓	SR↑	Ts↓	
1-assistant							
MCTS Planner w/o DM	33.3	7.0	29.9	1500.0	23.9	1500.0	29.0
LLM Planner w/o DM	27.5	14.6	39.7	1500.0	66.0	1236.2	44.4
2-assistant							
MCTS Planner w/o DM	34.0	6.9	27.1	1500.0	27.1	1500.0	29.4
LLM Planner w/o DM	25.0	15.2	55.6	1432.9	76.4	1060.2	52.3

no interaction with objects, it performs poorly in the other two tasks, where LLMs show difficulty tracking the task progress given only action history.

4.2 COMMUNITY INFLUENCE TASK

To further investigate agents’ planning and social capabilities in open-world settings, we introduce the *Community Influence Task*—a novel, open-ended social challenge in which two main agents compete to connect with and persuade other community members to form relationships with them. Due to differences in personality traits and social status, strategic planning is required to influence and shift member opinions over time.

Task Settings Each community contains two main agents and thirteen other members. The main agents must navigate the environment, locate potential members to connect with, and attempt to persuade them through dialogue. At the end of each day, every community member ranks their friendship level with the two main agents.

Observation and Action Spaces The observation and action spaces for all agents match those in the *Community Assistant Tasks*. Since this task focuses on social planning, both main agents are given access to the daily schedules of all members.

Experimental Settings and Evaluation Metrics We run experiments in five distinct communities. An agent is considered more effective at influencing others if it achieves a higher average friendship rank across all members.

Baseline We evaluate different LLMs as the planning backbone for the main agent. Given the daily schedules and character traits of community members, the main agent prompts the LLM to select the next member to visit, considering both spatial proximity and potential influence. After navigating to the target, the same LLM generates up to three rounds of conversation, conditioned on the main agent’s and the target member’s profiles. Once the conversation concludes, the main agent proceeds to the next selected target. Full prompt details are provided in the Appendix. [K](#).

Metrics We use two metrics: (1) *Average friendship-ranking wins (Win.)* — the average win rate in the friendship ranking across all community members at the end of the day. Higher values indicate that an agent is more effective at forming new connections and expanding its social influence. (2) *Conversion rate (Conv.)* — the proportion of originally non-supporting members who become friends with the agent by the end of the day.

Results As shown in Figure 4, the main agent with the o1 backbone achieves higher average friendship rankings and conversion rates than the gpt-4o backbone*, indicating greater ability to change members’ opinions in most communities. For example, in the Denver community, the agent, Yara Mbatha, persuaded Theo by leveraging their shared affiliation with the Denver Scholars Society and emphasizing that common bond. We also observe that when starting with a large advantage, the gpt-4o agent sometimes fails to gain additional supporters, primarily due to suboptimal target selection and less effective persuasion strategies. These results suggest that, even with advanced LLMs, there remains substantial room to improve embodied agents’ abilities in building social connections and exerting influence.

*We used GPT-4o-2024-11-20 and o1-2024-12-17 during experiments.

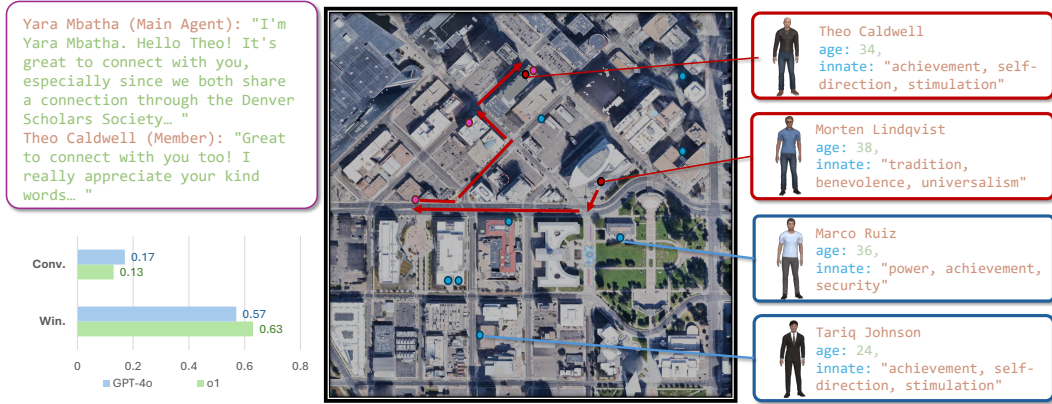


Figure 4: We evaluate baseline agents on the *Community Influence Task* across five communities. Results show that more powerful LLMs are better able to connect with and influence other members in the community.

Table 3:

Dialogue	Target Selection	Win.↑	Conv.↑
GPT-4o	GPT-4o	0.57	0.17
GPT-4o	o1	0.58	0.06
o1	GPT-4o	0.60	0.20
o1	o1	0.63	0.13

5 OPEN WORLD MULTI-ROBOT COOPERATION

In addition to high-level planning and interactions, we also explore low-level physics challenges in multi-agent, open-world settings. In this section, we introduce the *Community Robot Challenge*, which features scenarios where two heterogeneous robots cooperate to assist humans in open-world environments.

Task settings The *Community Robot Challenge* builds upon the *Carry* and *Deliver* categories from the *Community Assistant* tasks (Section 4.1), introducing physics-level collaboration in open-world environments. In this challenge, robots must cooperate to deliver an object to a destination or assist a human avatar by picking up and carrying personal items while following the avatar.

Robot Settings We use two robot assistants: a mobile manipulator—based on the Google robot model in MuJoCo (Todorov et al., 2012), augmented with one degree of freedom for forward translation and another for rotation about the z-axis—and a wheeled robot carrier with four degrees of freedom (one per wheel). In addition, Virtual Community supports quadruped and humanoid robots, which are described in the Appendix. D.

Observation and Control Spaces Observations include RGB-D images, segmentations, the base pose, and task-related information. The action space consists of 11 DoFs for the mobile manipulator (7 DoFs for the arm, 2 for the gripper, and 2 for locomotion) and 4 DoFs for the wheeled robot (1 for each wheel).

Baseline Pipeline We implement two baselines including *Heuristic* and *RL*. The heuristic baseline inherits the navigation module from baseline avatars in the *Community Assistant Tasks* 4.1. For navigation, robot computes a collision-free path with A* search. For manipulation, the robot solves for a feasible grasp pose with inverse kinematics, and plans and executes the grasp motion with RRT-Connect (Kuffner & LaValle, 2000). We also implemented a *VLA baseline*, which achieved near-zero performance. Full details are provided in the Appendix. F.6.

Table 4: **Detailed results of the Community Robot Challenge.** We report Success Rate (SR) and Time Consumed (Ts) for two community robot tasks averaged over 21 different scenes.

Method	Carry		Deliver		Avg SR↑
	SR↑	Ts↓	SR↑	Ts↓	
Heuristic	17.6	126.9	22.2	129.4	19.9
Heuristic w Oracle Grasp	23.5	124.4	50.0	131.2	36.8
RL w Oracle Grasp	19.0	149.7	42.9	168.1	31.0

Results According to the results in Table 4, all baselines achieve higher scores on the delivery task than on the carry task, highlighting the added difficulty of simultaneously manipulating objects and following a human in a dynamic open-world environment. Moreover, without using an oracle grasp, performance drops significantly, underscoring the challenge posed by the manipulation component in this task. The reinforcement learning baseline performs worse than the heuristic baseline, which uses inverse kinematics and RRTConnect to compute manipulation trajectories. This gap is because the classical planner explicitly solves for optimal paths in configuration space, whereas the RL agent must discover effective control sequences under sparse reward signals.

Table 5:

Method	Carry		Deliver		Avg SR↑
	SR↑	Ts↓	SR↑	Ts↓	
RL w/ Oracle Grasp	19.0	149.7	42.9	168.1	31.0
RL Decomposed w/ Oracle Grasp	19.0	149.0	47.6	172.1	33.3

6 CONCLUSION

We introduce Virtual Community, an open-world simulation platform for multi-agent embodied AI that supports scalable, simulation-ready generation of open-world scenes and agent communities, along with physically realistic simulation of multiple embodied avatars and robots. As an initial demonstration, we propose two novel open-world multi-agent challenges—the **Community Planning Challenge** and the **Community Robot Challenge**—each developed and evaluated using a variety of baseline methods. One limitation of this work is that the outdoor scenes are not modeled in sufficient detail to accurately reflect the physical and visual properties of real-world environments. We hope Virtual Community will advance embodied AI research toward embodied general intelligence capable of handling real-world complexities and coexisting with human communities.

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